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1 Appendix D: Neural Encoding Efficiency on Time-Series

1.1 Objective

Estimate information rate and rate-coding metrics from deterministic time-series and labels.

1.2 Planned Methods (src)

- `src/neural_encoding.py`
 - `information_rate_time_series(responses, dt_s, noise_std)`
 - `rate_coding_metrics(responses, labels)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_neural_encoding_analysis.py`
- Data: `output/data/neural_encoding.npz`
- Figure: `output/figures/neural_encoding_information_rate.png`

1.4 Figure

1.5 Equation References

- Information rate: see (??)
- Response time model: see (??)

1.6 Reproducibility

- Run: `python3 scripts/generate_neural_encoding_analysis.py`
- Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??

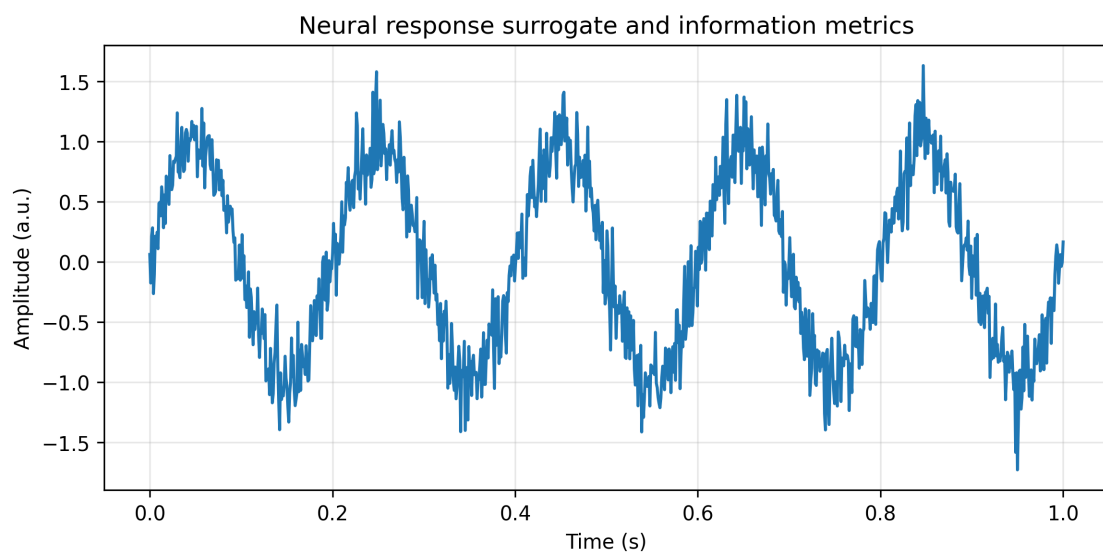


Figure 1: *Estimated information rate and classification metrics for IR-evoked response surrogates under controlled noise levels.*